Recitation 5

Feb 16, 2024

Ruiqi Lyu

Outline

1. Logistic Regression & Gradient Descent
2. Decision Tree
3. K-means

Logistic Regression

Logistic Regression is a Linear Classifier!

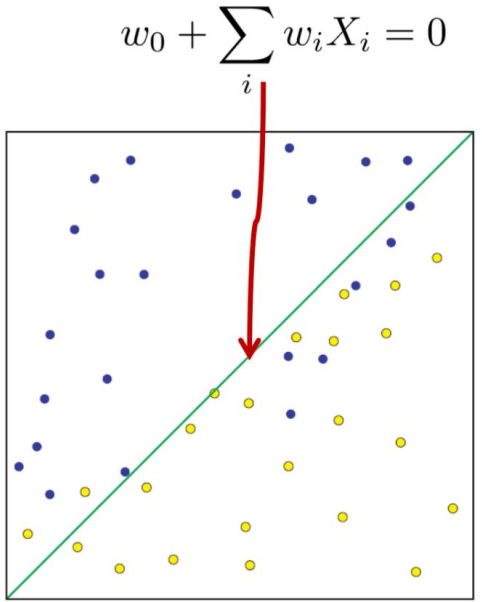
Assumes the following functional form for P(Y|X):

$$P(Y=1|X) = \frac{1}{1+\exp(-w_0 - \sum_i w_i X_i)}$$

Decision boundary:

$$\Rightarrow \frac{P(Y=1|X)}{P(Y=0|X)} = \exp(w_0 + \sum_i w_i X_i) \underset{0}{\overset{1}{\gtrless}} 1$$

$$\Rightarrow w_0 + \sum_i w_i X_i \underset{0}{\overset{1}{\gtrless}} 0$$



(Linear Decision Boundary)

Training Logistic Regression

How to learn the parameters w_0, w_1, … w_d? (d features)

Training Data $\{(X^{(j)}, Y^{(j)})\}_{j=1}^{n}$ $X^{(j)} = (X_1^{(j)}, \ldots, X_d^{(j)})$

Maximum Likelihood Estimates

$$\widehat{\mathbf{w}}_{MLE} = \arg\max_{\mathbf{w}} \prod_{j=1}^{n} P(X^{(j)}, Y^{(j)} \mid \mathbf{w})$$

But there is a problem …

Don't have a model for P(X) or P(X|Y) – only for P(Y|X)

Training Logistic Regression

How to learn the parameters w_0, w_1, … w_d? (d features)

Training Data $\{(X^{(j)}, Y^{(j)})\}_{j=1}^{n}$ $X^{(j)} = (X_1^{(j)}, \ldots, X_d^{(j)})$

Maximum (Conditional) Likelihood Estimates

$$\widehat{\mathbf{w}}_{MCLE} = \arg\max_{\mathbf{w}} \prod_{j=1}^{n} P(Y^{(j)} \mid X^{(j)}, \mathbf{w})$$

Discriminative philosophy – Don't waste effort learning P(X), focus on P(Y|X) – that's all that matters for classification!

Expressing Conditional log Likelihood

$$P(Y=1|X,w) = \frac{1}{1+\exp(-w_0 - \sum_i w_i X_i)}$$

$$P(Y=0|X,w) = \frac{1}{1+\exp(w_0 + \sum_i w_i X_i)}$$

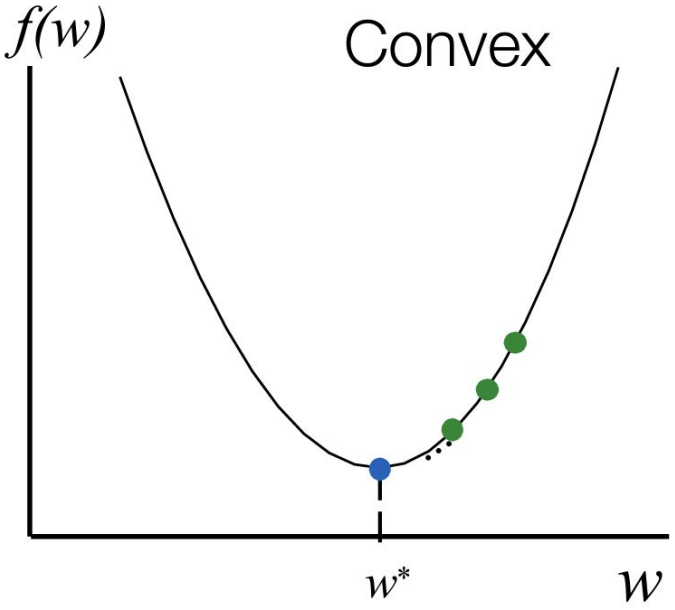
$$\begin{aligned} l(\mathbf{w}) &\equiv \ln \prod_j P(y^j|\mathbf{x}^j, \mathbf{w}) \\ &= \sum_j \left[y^j (w_0 + \sum_i^d w_i x_i^j) - \ln(1 + exp(w_0 + \sum_i^d w_i x_i^j)) \right] \end{aligned}$$

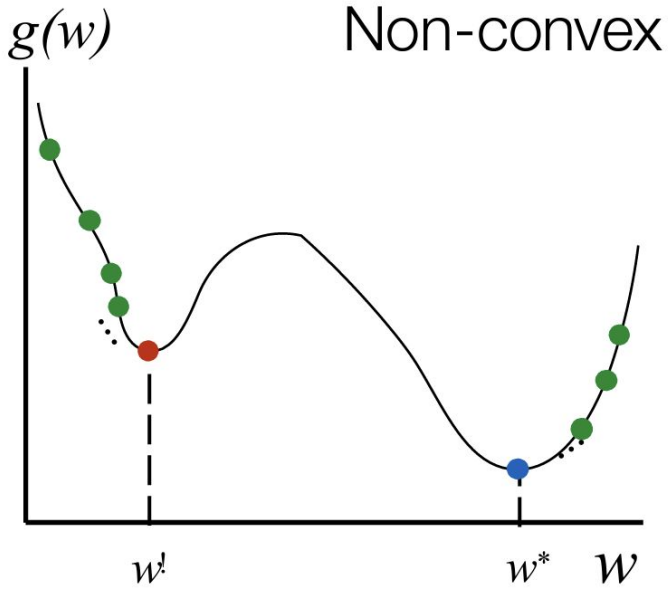
Good news: *l*(**w**) is concave function of **w !**

Bad news: no closed-form solution to maximize *l*(**w**) HW2!

Good news: can use iterative optimization methods (gradient ascent)

Gradient descent (with proper step size) converges to the global optimum for when minimizing a convex function.

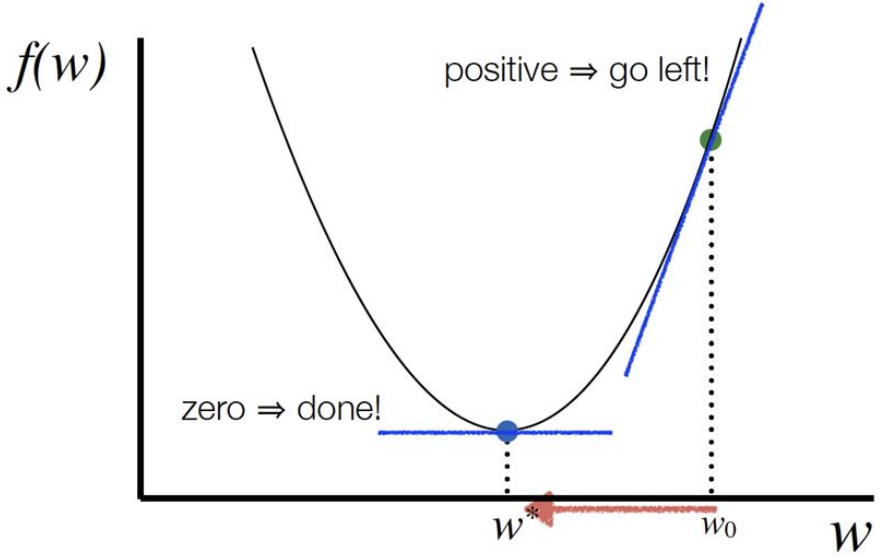


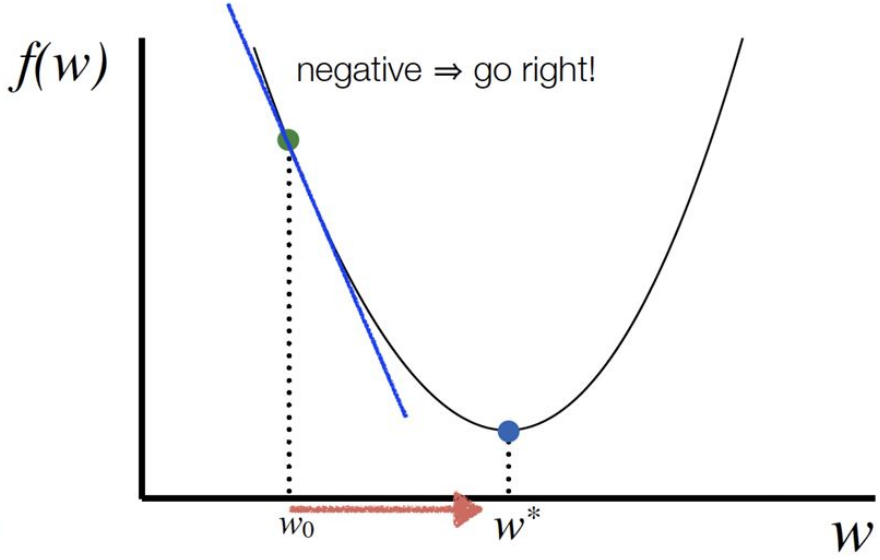


Any local minimum is also a global minimum.

Multiple local minima may exist.

Linear regression, ridge regression, and logistic regression are all convex!





We can only move in two directions

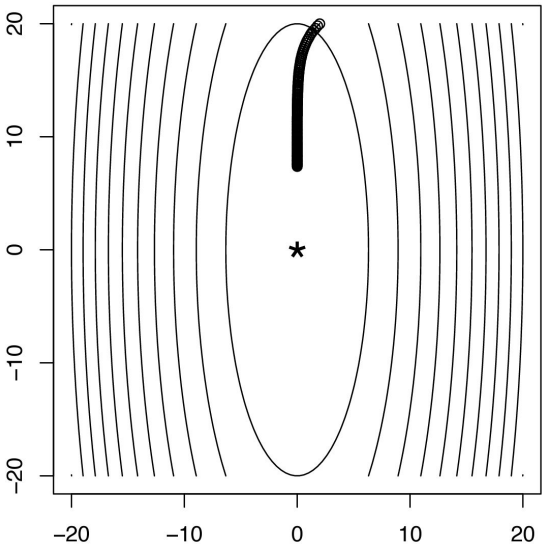
Negative slope is direction of descent!

Step Size

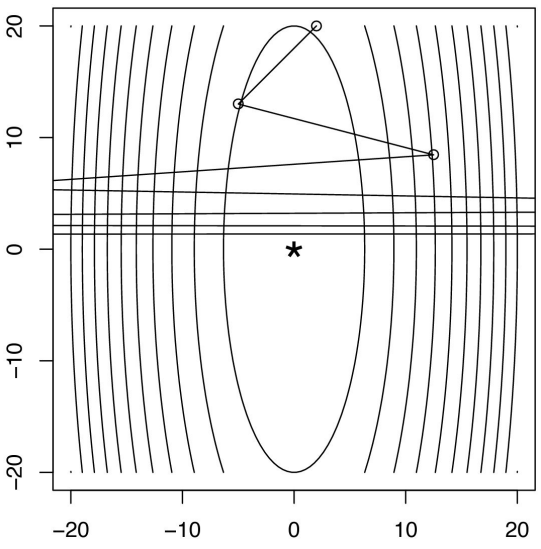
Update Rule: $w_{i+1} = w_i - \alpha_i \frac{df}{dw}(w_i)$

Negative Slope

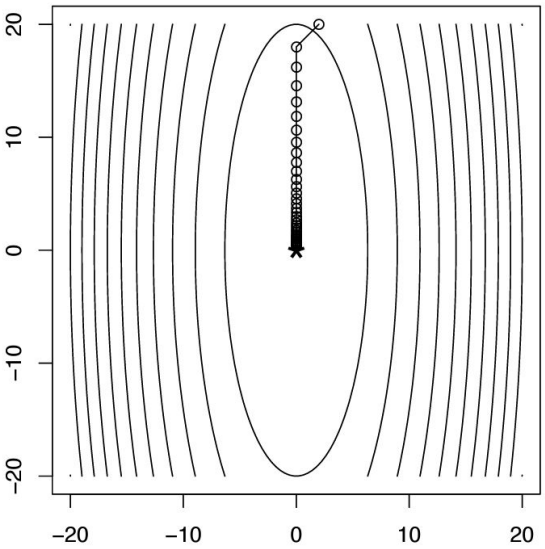
$$f(x) = (10{x_1}^2 + {x_2}^2/2)$$



100 steps



8 steps



40 steps

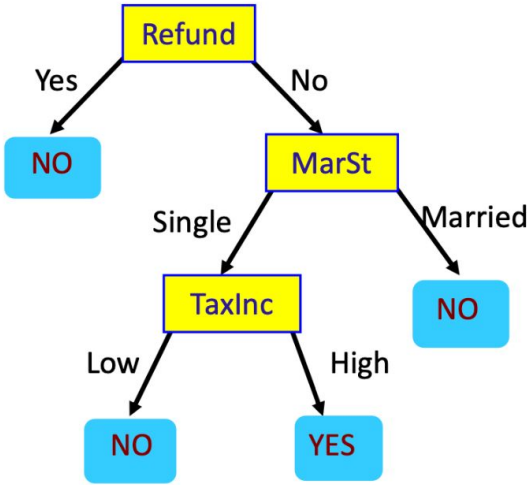
Decision Tree

How to learn a decision tree

- Top-down induction [ID3]

Main loop:

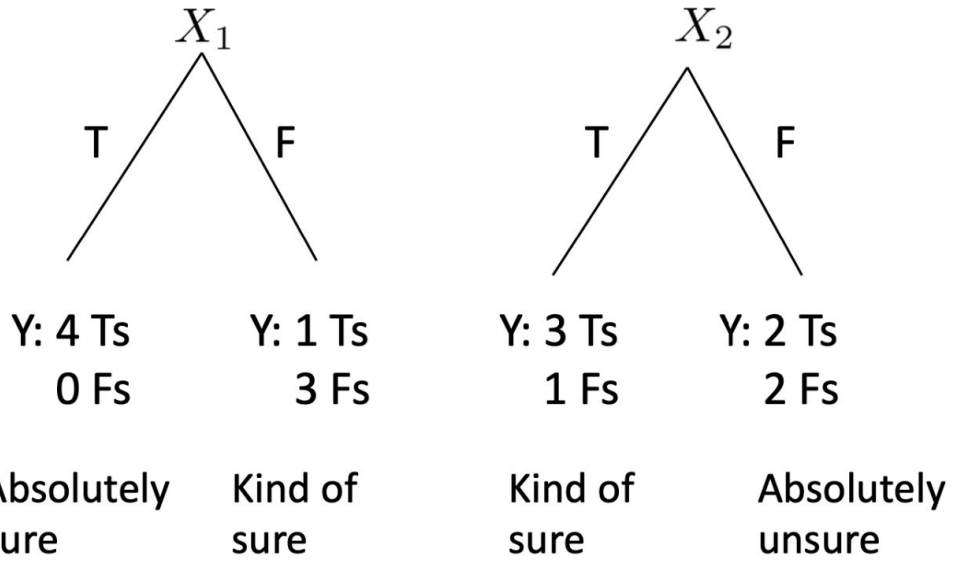
1. $X \leftarrow$ the "best" decision feature for next *node*
2. Assign X as decision feature for *node*
3. For each value of X, create new descendant of *node* (Discrete features)
4. Sort training examples to leaf nodes
5. If training examples perfectly classified, Then STOP, Else iterate over new leaf nodes
   (steps 1-5) after removing current feature



6. When all features exhausted, assign majority label to the leaf node

Which feature is best?

X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F
F	T	F
F	F	F



Good split if we are more certain about classification after split –

Uniform distribution of labels is bad

Which feature is best to split?

Pick the attribute/feature which yields maximum information gain:

$$\arg\max_i I(Y, X_i) = \arg\max_i [H(Y) - H(Y|X_i)]$$

$$= \arg\min_i H(Y|X_i)$$

Entropy of Y $\quad H(Y) = -\sum_y P(Y=y)\log_2 P(Y=y)$

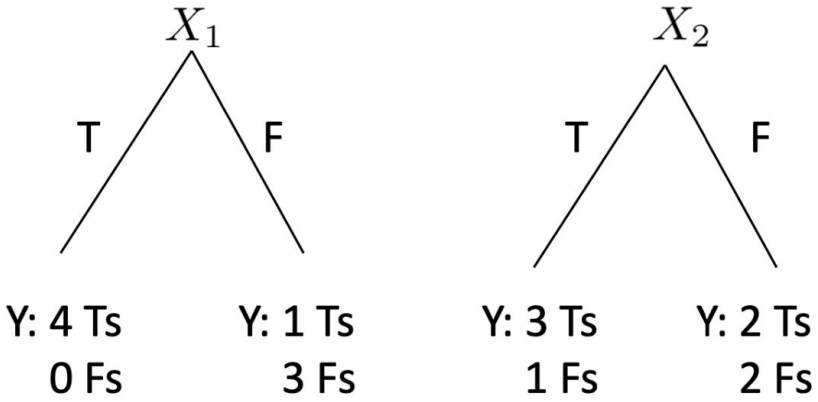
Conditional entropy of Y $\quad H(Y \mid X_i) = \sum_x P(X_i = x) H(Y \mid X_i = x)$

Feature which yields maximum reduction in entropy (uncertainty) provides maximum information about Y

Information Gain

$$H(Y \mid X_i) = -\sum_x P(X_i = x) \sum_y P(Y = y \mid X_i = x) \log_2 P(Y = y \mid X_i = x)$$

X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F
F	T	F
F	F	F



$$\widehat{H}(Y|X_1) = -\frac{1}{2}[1\log_2 1 + 0\log_2 0] - \frac{1}{2}[\frac{1}{4}\log_2\frac{1}{4} + \frac{3}{4}\log_2\frac{3}{4}]$$

$$\widehat{H}(Y|X_2) = -\frac{1}{2}[\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}] - \frac{1}{2}[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}]$$

> 0

K-means

K-means Clustering

Goal: Group the data into *k* clusters that minimize the sum of squared distances from each point to its respective cluster center (i.e. inter-cluster variance)

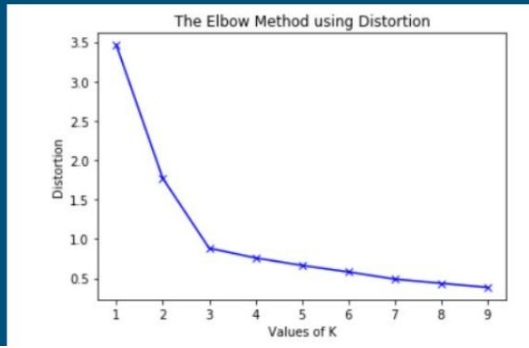
Objective Function:

$$Obj = \sum_{k} \sum_{x_i \in C_k} ||x_i - \mu_k||_2^2$$

Evaluating *k*:

- Choose the value *k* after which changes in the objective function begin to diminish

Source: Elbow Method



K-means Algorithm

1. Initialize *k* cluster centroids
2. Assign each data point to the nearest centroid
3. Estimate new cluster centroids
4. Repeat steps 2 and 3 until the stopping condition is met

Stopping Conditions

- All points remain in the same clusters
- Maximum number of iterations is reached
- Cluster centroids do not change

Demo codes for k-means

https://github.com/Rachel-Lyu/02620-S24-Machine-Learning-for-Scientists/blob/main/KMeans_Feb15.ipynb